KCE PATHWAYS · GRADES 4–5 · UNIT 6: VOICES OF CHANGE · WEEK 27 · MATH

The Planetarium Number Hunt — Teacher Lesson Plan & Answer Key

Learning objectives: Students explain patterns in the number of zeros when multiplying by powers of 10, write powers of 10 with whole-number exponents ($10^3 = 10 \times 10 \times 10 = 1{,}000$), multiply and divide whole numbers and decimals by powers of 10 by moving every digit left or right and filling empty places with placeholder zeros, and compare place values (a digit in one place is 10 times the digit to its right and 1/10 of the digit to its left).

Duration: 35–45 minutes
Materials: Computer or tablet with the lesson open · pencil · printed Pre & Post sheet

Lesson flow (as it appears on screen)

1. Slide 0 — Intro and name: meet Kiona (11), who is a helper at the Adler Planetarium in Chicago, Illinois, on the shore of Lake Michigan, with her cousin Kimani. Dr. Kostas, an astronomy educator, explains giant space numbers, Mr. Koji runs the star shows in the dome, and Kiona's friend Kirra visits with her own questions. The student types first and last name so results are sent. Read the scoring rules together: 2 tries per question, Pathway Points only for a correct first try, and the answer plus the reason appear after the 2nd miss. Typed number answers ignore capital letters, commas in large numbers, dollar signs, extra spaces and a final period.
2. Slide 1 — Warm-Up, Part 1: Expressions Review (4 pts, 2 each, Week 26 skill). If a student misses both warm-up slides, rework one expression step by step on scrap paper before moving on.
3. Slide 2 — Warm-Up, Part 2 (4 pts, 2 each): brackets and braces from the inside out, left-to-right order, or comparing without computing.
4. Slide 3 — Pencil Break (Pre): the student opens or prints the Pre/Post sheet (grade4-5/prepost/KCE_Math_PrePost_G4-5W27.pdf) and completes Page 1 (Pre-Lesson Warm-Up) off screen before any teaching.
5. Slide 4 — Why Astronomers Love Powers of 10: nonfiction connection about the Adler Planetarium, which opened in Chicago in 1930 as the first planetarium in the Western Hemisphere. Space numbers are huge: about 93,000,000 miles to the Sun (93×10^6), about 239,000 miles to the Moon on average, and light travels about 186,000 miles in one second. No questions.
6. Slide 5 — Picture It: Exponents Count the Tens. In 10^3, 10 is the base and 3 is the exponent: $10 \times 10 \times 10 = 1{,}000$. A table from 10^1 to 10^6 shows that the exponent equals the number of zeros. Watch out: 10^3 does NOT mean 10×3. With another factor: $4 \times 10^3 = 4{,}000$.
7. Slide 6 — Your Turn, Part 1: Exponents (8 pts, 4 each): a typed item (standard form, the missing exponent, or exponent form typed as 10^7 or 10^7) and a multiple-choice item (an equal power, a value, or the greatest value).
8. Slide 7 — Your Turn, Part 2 (4 pts): change each power of 10 to standard form, then add or multiply (star counts and a visitor count in expanded form).
9. Slide 8 — Teach: Multiplying by Powers of 10. Each digit moves one place to the left for every factor of 10: $3.6 \times 10^2 = 360$, with a placeholder zero in the empty ones place; $0.47 \times 10^3 = 470$; $45 \times 10^3 = 45{,}000$. Do not just add zeros to a decimal: 3.6×100 is not 3.600.

10. Slide 9 — Your Turn, Part 1: Multiplying by Powers of 10 (8 pts, 4 each): a typed product and a choice about the product or how many places the digits move.
11. Slide 10 — Your Turn, Part 2 (4 pts): a money or mass problem (stickers, rocket kits and posters, or a box of telescope lenses) that multiplies by 10^2 or 10^3, then finishes a second step.
12. Slide 11 — Teach: Dividing by Powers of 10. Each digit moves one place to the right for every factor of 10 and becomes 1/10 as large: $4{,}500 \div 10^2 = 45$; $72 \div 10^3 = 0.072$ (the placeholder zero in the tenths place matters). Sense check: the quotient must be smaller.
13. Slide 12 — Your Turn, Part 1: Dividing by Powers of 10 (8 pts, 4 each): a typed quotient and a choice (an equal expression, the greatest quotient, or a missing exponent).
14. Slide 13 — Your Turn, Part 2 (4 pts): divide by 10 or 10^2 twice in a row (a paper banner, cleaning kits shared by students, or glitter in jars and bags), watching for placeholder zeros.
15. Slide 14 — Vocabulary Round-Up: 8 flip cards (power of 10, base, exponent, standard form, place value, placeholder zero, factor, 1/10 as large).
16. Slide 15 — Your Turn: Using the Words (6 pts, 3 each): one multiple-choice and one typed vocabulary item.
17. Slide 16 — Teach: How Many Times as Large? Count the extra tens: 10^5 is 1,000 times as large as 10^2. The first 3 in 3,300 is 10 times the value of the second 3; the 2 in 0.2 is 10 times the value of the 2 in 0.02.
18. Slide 17 — Your Turn: How Many Times as Large? (6 pts, 3 each): compare two powers of 10 or two digits in one number, and find 1/10 of a decimal.
19. Slide 18 — Real World: Space Distances (4 pts): a different situation in each set (the Sun's distance as 93×10^n, how far light travels in 10^2 seconds, or the Moon's distance divided by 10^4 for a model in centimeters).
20. Slide 19 — Real World: Behind the Scenes (4 pts): 4.2×10^3 star cards shared by 10^2 classes, two meteorite masses in grams, or Kimani's claim that $10^3 \times 10^2 = 10^6$.
21. Slide 20 — Think It Through: Explain and Find the Mistake (10 pts, 5 each): a "which explanation is correct?" item and a "find the mistake" item. Ask the student to say why each wrong choice is wrong.
22. Slide 21 — Fix the Sentence, Part 1 (4 pts, 2 each): typed sentences checked exactly (quotation marks around dialogue, capital letters, commas with a city and state, and one grammar fix such as their/they're or to/too). Straight or curly quotes and extra spaces are OK. The corrected sentence appears after the 2nd miss.
23. Slide 22 — Fix the Sentence, Part 2 (4 pts, 2 each): two more sentences (a series, a renaming phrase, or an opening word with a comma, and a verb that agrees with its real subject). Both serial-comma and no-serial-comma forms of a list are accepted.
24. Slide 23 — Match-Up Round (10 pts): 5 powers of 10 matched to standard form, products and quotients of 3.7 or 37 matched to their values, or words (base, exponent, standard form, powers of 10, placeholder zero) matched to examples, 2 pts each on the first try.
25. Slide 24 — Speed Trail (bonus XP only): 3 quick power-of-10 questions against the student's own best time.
26. Slide 25 — Looking Back reflection (bonus XP only).
27. Slide 26 — Boss Battle: The Big Space Night (8 pts, 4 each): a different situation in each set (a gift shop order of 10^3 stickers and 10^2 posters, sunlight traveling for 5×10^2 seconds, or space dust grains magnified 10^3 and 10^2 times): choose the expression or first step, then finish the two-step answer.
28. Slide 27 — Pencil Break (Post): the student completes Page 2 (Post-Lesson Challenge) of the same sheet (grade4-5/prepost/KCE_Math_PrePost_G4-5W27.pdf), plus the ★ Challenge if time allows, before seeing the score.
29. Slide 28 — Results (Planetarium Day Complete!): the score out of 100 is sent to KingCretot Experience. 85 or higher shows "Adler Planetarium Power Master!" with the congratulations message "Your space numbers passed every check at the Adler Planetarium!" and a big confetti burst; below 85 shows "Good Work at the Planetarium!" with encouragement ("You are getting closer every time."), and the next attempt brings a fresh question set (sets rotate 1, 2, 3). Both messages remind the student to finish the Post-Lesson Challenge sheet.

Lesson answer key with reasoning

Every scored item the student sees, in order, across all three rotating question sets. Each item allows 2 tries and earns Pathway Points only on a correct first try. The "why" is the same reasoning shown on screen after a second missed try.

WARM-UP: EXPRESSIONS REVIEW (2 POINTS EACH)

ID	Question	Answer	Why (explain to student)
r1_s0	Set 1: Order of operations (Week 26): evaluate 6 + 9 × 3. Value:	**33**	Multiply first: 9 × 3 = 27. Then add: 6 + 27 = 33.
r2_s0	Set 1: Writing expressions (Week 26): which expression means “add 14 and 6, then divide by 5”? [(14 + 6) ÷ 5 / 14 + 6 ÷ 5 / 14 ÷ 5 + 6 / 5 ÷ (14 + 6)]	**(14 + 6) ÷ 5**	The words say add first, so 14 + 6 goes in parentheses before dividing by 5.
r3_s0	Set 1: Grouping symbols (Week 26): evaluate 2 × [30 − (8 + 7)]. Value:	**30**	Parentheses: 8 + 7 = 15. Brackets: 30 − 15 = 15. Then 2 × 15 = 30.
r4_s0	Set 1: Order of operations (Week 26): evaluate 45 − 15 + 10. Value:	**40**	Add and subtract from left to right: 45 − 15 = 30, then 30 + 10 = 40.
r1_s1	Set 2: Order of operations (Week 26): evaluate 64 ÷ 8 × 2. Value:	**16**	Divide and multiply from left to right: 64 ÷ 8 = 8, then 8 × 2 = 16.
r2_s1	Set 2: Parentheses (Week 26): where do the parentheses go so that 5 + 3 × 6 = 48? [(5 + 3) × 6 / 5 + (3 × 6) / (5) + 3 × 6 / 5 + 3 × (6)]	**(5 + 3) × 6**	(5 + 3) × 6 = 8 × 6 = 48. The other choices multiply 3 × 6 first and give 23.
r3_s1	Set 2: Grouping symbols (Week 26): evaluate {50 − [3 × (4 + 6)]} ÷ 5. Value:	**4**	Parentheses: 4 + 6 = 10. Brackets: 3 × 10 = 30. Braces: 50 − 30 = 20. Then 20 ÷ 5 = 4.
r4_s1	Set 2: Comparing (Week 26): without computing, 6 × (517 + 88) is how many times as large as 517 + 88? Times as large:	**6** (also accepted: 6 times, six)	The whole sum is multiplied by 6, so it is 6 times as large.
r1_s2	Set 3: Order of operations (Week 26): evaluate (9 + 11) × 4 − 7. Value:	**73**	Parentheses: 9 + 11 = 20. Multiply: 20 × 4 = 80. Subtract: 80 − 7 = 73.
r2_s2	Set 3: Comparing (Week 26): which expression is NOT four times as large as 230 + 16? [4 × 230 + 16 / 4 × (230 + 16) / (230 + 16) × 4 / (230 + 16) + (230 + 16) + (230 + 16) + (230 + 16)]	**4 × 230 + 16**	In 4 × 230 + 16, only 230 is multiplied by 4. The others multiply the whole sum by 4.
r3_s2	Set 3: Grouping symbols (Week 26): evaluate 3 × [2 + (12 − 5)]. Value:	**27**	Parentheses: 12 − 5 = 7. Brackets: 2 + 7 = 9. Then 3 × 9 = 27.
r4_s2	Set 3: Order of operations (Week 26): evaluate 7 × 6 − 24 ÷ 3. Value:	**34**	Multiply and divide first: 7 × 6 = 42 and 24 ÷ 3 = 8. Then subtract: 42 − 8 = 34.

YOUR TURN: EXPONENTS (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
l1_s0	Set 1: Write 10^4 in standard form. Standard form:	**10,000**	10^4 = 10 × 10 × 10 × 10 = 10,000, a 1 followed by 4 zeros.
l2_s0	Set 1: Which is equal to 10 × 10 × 10 × 10 × 10? [10^5 / 10 × 5 / 5^{10} / 10^4]	**10^5**	Five tens are multiplied, so the base is 10 and the exponent is 5: 10^5 = 100,000. 10 × 5 is only 50.

ID	Question	Answer	Why (explain to student)
l3_s0	Set 1: Two steps: Mr. Koji's star show puts 3 × 10^4 stars on the dome. Then he adds 5 × 10^3 more stars. How many stars are on the dome in all? Stars:	**35,000** (also accepted: 35,000 stars)	3 × 10^4 = 30,000 and 5 × 10^3 = 5,000. 30,000 + 5,000 = 35,000 stars.
l1_s1	Set 2: 1,000,000 = 10 to what power? Type the exponent. Exponent:	**6** (also accepted: six)	1,000,000 has 6 zeros, so it equals 10 × 10 × 10 × 10 × 10 × 10 = 10^6. The exponent is 6.
l2_s1	Set 2: What is the value of 10^2? [100 / 20 / 12 / 1,000]	**100**	10^2 = 10 × 10 = 100. It is not 10 × 2 = 20.
l3_s1	Set 2: Two steps: Kimani writes a visitor count as 7 × 10^3 + 5 × 10^2 + 4 × 10. Write the number in standard form. Standard form:	**7,540**	7 × 1,000 = 7,000, 5 × 100 = 500, and 4 × 10 = 40. 7,000 + 500 + 40 = 7,540.
l1_s2	Set 3: Write 10 × 10 × 10 × 10 × 10 × 10 × 10 in exponent form. (Type 10^ and then the exponent.)	**10^7** (also accepted: 10^7, 10 to the 7th power, 10 to the seventh power, 10 to the 7th, 10 to the seventh)	Seven tens are multiplied, so the exponent is 7: 10^7.
l2_s2	Set 3: Which has the GREATEST value? [10^5 / 5 × 10^4 / 99,999 / 9 × 10^3]	**10^5**	10^5 = 100,000. 5 × 10^4 = 50,000, 99,999 is 1 less than 100,000, and 9 × 10^3 = 9,000.
l3_s2	Set 3: Two steps: Dr. Kostas's first star map shows 6 × 10^3 stars. A second map shows 10 times as many stars. How many stars does the second map show? Stars:	**60,000** (also accepted: 60,000 stars)	6 × 10^3 = 6,000. Ten times as many: 6,000 × 10 = 60,000 stars (which is 6 × 10^4).

YOUR TURN: MULTIPLYING BY POWERS OF 10 (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
k1_s0	Set 1: Multiply: 5.8 × 10^2 Product:	**580**	10^2 means move each digit 2 places to the left: 5.8 → 58 → 580.
k2_s0	Set 1: Which number equals 2.4 × 10^3? [2,400 / 240 / 24,000 / 2.4000]	**2,400**	10^3 moves each digit 3 places to the left: 2.4 → 24 → 240 → 2,400. Writing 2.4000 adds zeros after the decimal point, which does not change the value.
k3_s0	Set 1: Two steps: at the gift shop, Kiona buys 10^2 star stickers that cost $0.15 each. She pays with a $20 bill. How much change does she get? Dollars:	**5** (also accepted: 5 dollars, 5 dollar, 5.00, 5.00 dollars, 5.00 dollar)	10^2 stickers is 100 stickers: 0.15 × 100 = $15. Change: 20 − 15 = $5.
k1_s1	Set 2: Multiply: 0.36 × 10^3 Product:	**360**	10^3 moves each digit 3 places to the left: 0.36 → 3.6 → 36 → 360.
k2_s1	Set 2: In 0.6 × 10^4, how many places does each digit move to the left? [4 / 1 / 6 / 3]	**4**	The exponent 4 means four factors of 10, so each digit moves 4 places: 0.6 × 10^4 = 6,000.
k3_s1	Set 2: Two steps: the planetarium orders 10^3 model rocket kits at $8.50 each and 10^2 posters at $3.20 each. What is the total cost? Dollars:	**8,820** (also accepted: 8,820 dollars, 8,820 dollar, 8,820.00, 8,820.00 dollars, 8,820.00 dollar)	Kits: 8.50 × 1,000 = $8,500. Posters: 3.20 × 100 = $320. Total: 8,500 + 320 = $8,820.

ID	Question	Answer	Why (explain to student)
k1_s2	Set 3: Multiply: 7.09 × 10^2 Product:	**709**	10^2 moves each digit 2 places to the left: 7.09 → 70.9 → 709. The 0 in the tenths place becomes the 0 in the tens place.
k2_s2	Set 3: Which product equals 48,000? [4.8 × 10^4 / 4.8 × 10^3 / 48 × 10^4 / 0.48 × 10^3]	**4.8 × 10^4**	4.8 × 10^4 = 48,000. The others: 4.8 × 10^3 = 4,800, 48 × 10^4 = 480,000, and 0.48 × 10^3 = 480.
k3_s2	Set 3: Two steps: each lens for a classroom telescope has a mass of 0.035 kilogram. A box holds 10^3 lenses, and the empty box has a mass of 2.5 kilograms. What is the mass of the full box? Kilograms:	**37.5** (also accepted: 37.5 kilograms, 37.5 kilogram, 37.5 kg)	Lenses: 0.035 × 1,000 = 35 kilograms. Add the box: 35 + 2.5 = 37.5 kilograms.

YOUR TURN: DIVIDING BY POWERS OF 10 (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
e1_s0	Set 1: Divide: 6,300 ÷ 10^2 Quotient:	**63**	10^2 moves each digit 2 places to the right: 6,300 → 630 → 63.
e2_s0	Set 1: Which expression equals 0.52? [520 ÷ 10^3 / 520 ÷ 10^2 / 52 ÷ 10^3 / 5.2 × 10]	**520 ÷ 10^3**	520 ÷ 10^3 moves each digit 3 places to the right: 0.52. The others: 520 ÷ 10^2 = 5.2, 52 ÷ 10^3 = 0.052, and 5.2 × 10 = 52.
e3_s0	Set 1: Two steps: Dr. Kostas cuts a 4.5-meter paper banner into 10 equal pieces. Then he cuts one piece into 10 equal smaller pieces. How long is one smaller piece? Meters:	**0.045** (also accepted: 0.045 meters, 0.045 meter, 0.045 m)	First cut: 4.5 ÷ 10 = 0.45 meter. Second cut: 0.45 ÷ 10 = 0.045 meter. In all, that is 4.5 ÷ 10^2.
e1_s1	Set 2: Divide: 85 ÷ 10^3 Quotient:	**0.085**	10^3 moves each digit 3 places to the right: 85 → 8.5 → 0.85 → 0.085. A placeholder zero holds the tenths place.
e2_s1	Set 2: Which quotient is the GREATEST? [4,000 ÷ 10^2 / 4,000 ÷ 10^3 / 400 ÷ 10^3 / 40 ÷ 10^3]	**4,000 ÷ 10^2**	4,000 ÷ 10^2 = 40. The others: 4,000 ÷ 10^3 = 4, 400 ÷ 10^3 = 0.4, and 40 ÷ 10^3 = 0.04.
e3_s1	Set 2: Two steps: Kimani's class spends $1,280 on 10^2 telescope cleaning kits that all cost the same. Each kit's cost is shared equally by 10 students. How much does each student pay? Dollars:	**1.28** (also accepted: 1.28 dollars, 1.28 dollar)	One kit: 1,280 ÷ 100 = $12.80. Each student: 12.80 ÷ 10 = $1.28.
e1_s2	Set 3: Divide: 9.4 ÷ 10^2 Quotient:	**0.094**	10^2 moves each digit 2 places to the right: 9.4 → 0.94 → 0.094.
e2_s2	Set 3: 27 ÷ 10^n = 0.027. What is n? [3 / 2 / 4 / 1]	**3**	The 7 moved from the ones place to the thousandths place, which is 3 places to the right, so n = 3.
e3_s2	Set 3: Two steps: Mr. Koji has 7,200 grams of star-show glitter. He shares it equally among 10^2 jars, then shares each jar equally among 10 bags. How many grams go in each bag? Grams:	**7.2** (also accepted: 7.2 grams, 7.2 gram, 7.2 g)	Each jar: 7,200 ÷ 100 = 72 grams. Each bag: 72 ÷ 10 = 7.2 grams.

YOUR TURN: USING THE WORDS (3 POINTS EACH)

ID	Question	Answer	Why (explain to student)
v1_s0	Set 1: From the lesson: "In 10^3, the 10 is the base and the 3 is the exponent." In 10^5, what is the exponent? [5 / 10 / 50 / 100,000]	**5**	The exponent is the small raised number. In 10^5 it is 5, and the base is 10.
v2_s0	Set 1: Type your answer: a number written with digits, like 4,000, is in ___ form. (one word)	**standard**	Standard form uses digits. 4×10^3 is the same number written another way.
v1_s1	Set 2: From the lesson: "A power of 10 is a number like 10, 100, or 1,000 that you get by multiplying tens together." Which number is a power of 10? [10,000 / 2,000 / 110 / 1,010]	**10,000**	10,000 = 10 × 10 × 10 × 10. The others cannot be made by multiplying only tens.
v2_s1	Set 2: Type your answer: in 10^4, the number 10 is called the ___. (one word)	**base** (also accepted: the base)	The base is the number used as a factor. The exponent, 4, tells how many times.
v1_s2	Set 3: From the lesson: "A placeholder zero holds an empty place so the other digits stay in the right places." In 0.072, why is the 0 right after the decimal point needed? [It holds the tenths place, so the 7 stays in the hundredths place / It makes the number bigger / It is not needed, because 0.072 = 0.72 / It shows the number is less than zero]	**It holds the tenths place, so the 7 stays in the hundredths place**	Without the placeholder zero, the 7 would slide into the tenths place, and 0.72 is ten times as large as 0.072.
v2_s2	Set 3: Type your answer: in 10^3, the small raised 3 is called the ___. (one word)	**exponent** (also accepted: the exponent)	The exponent tells how many times 10 is used as a factor.

YOUR TURN: HOW MANY TIMES AS LARGE? (3 POINTS EACH)

ID	Question	Answer	Why (explain to student)
t1_s0	Set 1: 10^6 is how many times as large as 10^4? Times as large:	**100** (also accepted: 100 times)	10^6 has 2 more factors of 10 than 10^4: 10 × 10 = 100 times as large.
t2_s0	Set 1: In 0.44, the first 4 is how many times the value of the second 4? [10 times / 1/10 times / 100 times / 4 times]	**10 times**	The first 4 is in the tenths place and the second is in the hundredths place, one place to the right. A place is 10 times the place to its right.
t1_s1	Set 2: In 5,555, the first 5 is how many times the value of the last 5? Times as large:	**1,000** (also accepted: 1,000 times)	The first 5 is worth 5,000 and the last 5 is worth 5. That is 3 places apart: 10 × 10 × 10 = 1,000 times.
t2_s1	Set 2: Which number is 1/10 of 3.6? [0.36 / 36 / 0.036 / 3.06]	**0.36**	Taking 1/10 is dividing by 10: every digit moves one place to the right, so 3.6 becomes 0.36.
t1_s2	Set 3: 8×10^5 is how many times as large as 8×10^2? Times as large:	**1,000** (also accepted: 1,000 times)	Both have 8 as a factor. 10^5 has 3 more factors of 10 than 10^2, so it is 1,000 times as large.
t2_s2	Set 3: Which statement is true? [10^4 is 10 times as large as 10^3 / 10^4 is 4 times as large as 10^3 / 10^4 = 40 / 10^4 is 1,000 times as large as 10^3]	**10^4 is 10 times as large as 10^3**	10^4 = 10,000 and 10^3 = 1,000. 10,000 is 10 times 1,000.

REAL WORLD: SPACE DISTANCES (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
w1_s0	Set 1: *The Situation Earth is about 93,000,000 miles from the Sun. Dr. Kostas writes this distance on the whiteboard as 93×10^n miles.* Two steps: what number is n? n =	**6** (also accepted: six)	93,000,000 = 93 × 1,000,000, and 1,000,000 has 6 zeros, so it is 10^6. n = 6.
w1_s1	Set 2: *The Situation Light travels about 186,000 miles in one second. Kirra wants to know how far light travels in 10^2 seconds.* Two steps: about how far does light travel in 10^2 seconds? [18,600,000 miles / 1,860,000 miles / 186,000,000 miles / 18,600 miles]	**18,600,000 miles**	10^2 seconds = 100 seconds. 186,000 × 100 moves every digit 2 places to the left: 18,600,000 miles.
w1_s2	Set 3: *The Situation The Moon is about 239,000 miles from Earth on average. A model in the exhibit hall divides real distances in miles by 10^4 and shows the answer in centimeters.* Two steps: how many centimeters apart are the model Earth and the model Moon? Centimeters:	**23.9** (also accepted: 23.9 centimeters, 23.9 centimeter, 23.9 cm)	10^4 = 10,000. 239,000 ÷ 10,000 moves every digit 4 places to the right: 23.9 centimeters.

REAL WORLD: BEHIND THE SCENES (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
w2_s0	Set 1: *The Situation Dr. Kostas has 4.2×10^3 star cards to give away. He shares them equally among 10^2 visiting classes.* Two steps: how many star cards does each class get? [42 / 420 / 4.2 / 4,200]	**42**	4.2×10^3 = 4,200 cards. 4,200 ÷ 10^2 = 4,200 ÷ 100 = 42 cards for each class.
w2_s1	Set 2: *The Situation Two meteorite pieces sit in a display case. One has a mass of 0.85 kilogram and the other has a mass of 1.4 kilograms. One kilogram is 10^3 grams.* Two steps: what is the total mass of the two pieces in grams? Grams:	**2,250** (also accepted: 2,250 grams, 2,250 gram, 2,250 g)	Total: 0.85 + 1.4 = 2.25 kilograms. Change to grams: 2.25×10^3 = 2,250 grams.
w2_s2	Set 3: *The Situation Kimani writes in his notebook: $10^3 \times 10^2 = 10^6$, because 3 × 2 = 6.* Two steps: is Kimani right? [No: 1,000 × 100 = 100,000, which is 10^5 / Yes: 3 × 2 = 6, so it is 10^6 / No: 1,000 × 100 = 1,100, which is 10^4 / Yes: 1,000 × 100 = 1,000,000]	**No: 1,000 × 100 = 100,000, which is 10^5**	Write each power in standard form: 1,000 × 100 = 100,000. That is five tens multiplied, 10^5. Three tens and two more tens make five tens.

THINK IT THROUGH: EXPLAIN AND FIND THE MISTAKE (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
x1_s0	Set 1: Which explanation is correct for 10^3? [10 × 10 × 10 = 1,000, because the exponent tells how many tens are multiplied / 10 × 3 = 30, because the exponent is a factor / 10 + 10 + 10 = 30, because there are three tens / 10 × 10 × 10 × 10 = 10,000, because the 3 means three multiplication signs]	**10 × 10 × 10 = 1,000, because the exponent tells how many tens are multiplied**	An exponent counts factors of the base: 10^3 = 10 × 10 × 10 = 1,000.
x2_s0	Set 1: Find the mistake: Kimani says 4.5 × 10^2 = 4.500, because you add two zeros. What should he fix? [Each digit moves 2 places to the left: 4.5 × 100 = 450 / Nothing. 4.500 is correct / It should be 45 / It should be 4,500]	**Each digit moves 2 places to the left: 4.5 × 100 = 450**	Zeros after a decimal do not change its value: 4.500 = 4.5. Multiplying by 100 moves each digit 2 places to the left, which gives 450.
x1_s1	Set 2: Which explanation is correct for 0.3 × 10 = 3? [Each digit becomes 10 times as large, so 3 tenths become 3 ones / Add a zero to the end: 0.30 / The 10 cancels out the 0 in 0.3 / Multiplying by 10 always makes a number end in zero]	**Each digit becomes 10 times as large, so 3 tenths become 3 ones**	Multiplying by 10 moves each digit one place to the left: 3 tenths × 10 = 3 ones. 0.30 is still only 0.3.
x2_s1	Set 2: Find the mistake: Kiona says 56 ÷ 10^3 = 0.56. What should she fix? [Each digit moves 3 places to the right, so the answer is 0.056 / Nothing. 0.56 is correct / It should be 56,000 / It should be 5.6]	**Each digit moves 3 places to the right, so the answer is 0.056**	56 ÷ 10^3 moves each digit 3 places: 5.6, then 0.56, then 0.056. She moved only 2 places, which is dividing by 10^2.
x1_s2	Set 3: Which explanation is correct? Is 2 × 10^4 the same as 20,000? [Yes: 10^4 = 10,000, and 2 × 10,000 = 20,000 / No: it is 2 × 40 = 80 / No: it is the same as 20^4 / Yes: because 2 + 4 = 6 zeros]	**Yes: 10^4 = 10,000, and 2 × 10,000 = 20,000**	10^4 = 10,000, and 2 × 10,000 = 20,000. That has 4 zeros, not 6.
x2_s2	Set 3: Find the mistake: Kirra says 10^5 has 5 digits. What should she fix? [10^5 = 100,000, which has 5 zeros but 6 digits / Nothing. It has 5 digits / It has 4 digits / It has 50 digits]	**10^5 = 100,000, which has 5 zeros but 6 digits**	10^5 = 100,000: a 1 followed by 5 zeros, so 6 digits in all.

FIX THE SENTENCE (2 POINTS EACH)

ID	Question	Answer	Why (explain to student)
s1_s0	Set 1: Fix this sentence: kiona asked, how big is the dome	**Kiona asked, "How big is the dome?"**	Capitalize the name Kiona, put quotation marks around her exact words, capitalize the first word of the quote, and put the question mark inside the closing quotation mark.
s2_s0	Set 1: Fix this sentence: The adler planetarium is next to lake michigan in chicago, illinois.	**The Adler Planetarium is next to Lake Michigan in Chicago, Illinois.**	Names of buildings, lakes, cities, and states are proper nouns, so capitalize Adler Planetarium, Lake Michigan, Chicago, and Illinois.
s3_s0	Set 1: Fix this sentence: We saw planets stars and comets in the show.	**We saw planets, stars, and comets in the show.** (also accepted: We saw planets, stars and comets in the show.)	Use commas to separate the items in a series: planets, stars, and comets.

ID	Question	Answer	Why (explain to student)
s4_s0	Set 1: Fix this sentence: each of the exhibits have a star map	**Each of the exhibits has a star map.**	Capitalize the first word. The subject is each (one exhibit at a time), so use has, not have. End with a period.
s1_s1	Set 2: Fix this sentence: dr. kostas said, look up at the dome	**Dr. Kostas said, "Look up at the dome."** (also accepted: Dr. Kostas said, "Look up at the dome!")	Capitalize Dr. and the name Kostas, put quotation marks around his exact words, capitalize the first word of the quote, and put the end mark inside the closing quotation mark.
s2_s1	Set 2: Fix this sentence: The visitors put on they're special glasses.	**The visitors put on their special glasses.**	Their shows ownership: the glasses belong to the visitors. They're means they are.
s3_s1	Set 2: Fix this sentence: Mr. Koji the show operator dimmed the lights.	**Mr. Koji, the show operator, dimmed the lights.**	The show operator describes Mr. Koji. Put a comma before and after a phrase like this that renames the person.
s4_s1	Set 2: Fix this sentence: the stars in the show is very bright	**The stars in the show are very bright.**	Capitalize the first word. The subject is stars (plural), not show, so use are, not is. End with a period.
s1_s2	Set 3: Fix this sentence: kimani shouted, that star is huge	**Kimani shouted, "That star is huge!"** (also accepted: Kimani shouted, "That star is huge.")	Capitalize the name Kimani, put quotation marks around his exact words, capitalize the first word of the quote, and put the end mark inside the closing quotation mark.
s2_s2	Set 3: Fix this sentence: Kirra wants to see the comet to.	**Kirra wants to see the comet too.**	Too means also: Kirra wants to see the comet, also. The first to (to see) is correct.
s3_s2	Set 3: Fix this sentence: Well I think the answer is one thousand.	**Well, I think the answer is one thousand.**	Put a comma after an opening word like Yes, No, or Well.
s4_s2	Set 3: Fix this sentence: the lenses in the telescope is clean	**The lenses in the telescope are clean.**	Capitalize the first word. The subject is lenses (plural), not telescope, so use are, not is. End with a period.

MATCH-UP ROUND (2 POINTS PER FIRST-TRY MATCH)

ID	Question	Answer	Why (explain to student)
match_s0_1	Set 1: Match 10^2	**100**	2 points if matched on the first try.
match_s0_2	Set 1: Match 10^3	**1,000**	2 points if matched on the first try.
match_s0_3	Set 1: Match 10^4	**10,000**	2 points if matched on the first try.
match_s0_4	Set 1: Match 10^5	**100,000**	2 points if matched on the first try.
match_s0_5	Set 1: Match 10^6	**1,000,000**	2 points if matched on the first try.
match_s1_1	Set 2: Match 3.7×10^2	**370**	2 points if matched on the first try.
match_s1_2	Set 2: Match 3.7×10	**37**	2 points if matched on the first try.
match_s1_3	Set 2: Match 3.7×10^3	**3,700**	2 points if matched on the first try.
match_s1_4	Set 2: Match $37 \div 10^2$	**0.37**	2 points if matched on the first try.
match_s1_5	Set 2: Match $37 \div 10^3$	**0.037**	2 points if matched on the first try.

ID	Question	Answer	Why (explain to student)
match_s2_1	Set 3: Match the 10 in 10^3	**base**	2 points if matched on the first try.
match_s2_2	Set 3: Match the 3 in 10^3	**exponent**	2 points if matched on the first try.
match_s2_3	Set 3: Match 4,000 written with digits	**standard form**	2 points if matched on the first try.
match_s2_4	Set 3: Match 100 or 10,000	**powers of 10**	2 points if matched on the first try.
match_s2_5	Set 3: Match a zero that holds an empty place	**placeholder zero**	2 points if matched on the first try.

BOSS BATTLE: THE BIG SPACE NIGHT (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
b1_s0	Set 1: *Kiona Stocks the Gift Shop For a big space night, the gift shop orders 10^3 glow-in-the-dark star stickers that cost $0.75 each and 10^2 posters that cost $4.20 each.* Which expression gives the total cost? [$10^3 \times 0.75 + 10^2 \times 4.20$ / $(10^3 + 10^2) \times 0.75$ / $10^3 \times (0.75 + 4.20)$ / $10^3 + 0.75 \times 10^2 + 4.20$]	**$10^3 \times 0.75 + 10^2 \times 4.20$**	Each item has its own count and price: stickers $10^3 \times 0.75$, posters $10^2 \times 4.20$. Add the two costs.
b2_s0	Set 1: Two steps: what is the total cost? Dollars:	**1,170** (also accepted: 1,170 dollars, 1,170 dollar, 1,170.00, 1,170.00 dollars, 1,170.00 dollar)	Stickers: 0.75 × 1,000 = $750. Posters: 4.20 × 100 = $420. Total: 750 + 420 = $1,170.
b1_s1	Set 2: *Dr. Kostas's Sunlight Puzzle Light travels about 186,000 miles in one second. Sunlight takes about 500 seconds, a little more than 8 minutes, to reach Earth. Dr. Kostas writes 500 as 5×10^2.* Which expression shows about how far sunlight travels to reach Earth? [$186{,}000 \times 5 \times 10^2$ / 186,000 + 500 / 186,000 ÷ 500 / $186 \times 5 \times 10^2$]	**$186{,}000 \times 5 \times 10^2$**	Distance = miles each second × number of seconds: 186,000 × 500, and $500 = 5 \times 10^2$.
b2_s1	Set 2: Two steps: about how many miles does sunlight travel to reach Earth? Miles:	**93,000,000** (also accepted: 93,000,000 miles, 93,000,000 mile, 93 million, 93 million miles)	186,000 × 5 = 930,000. Then × 10^2 moves every digit 2 places to the left: 93,000,000 miles. That matches the distance to the Sun!
b1_s2	Set 3: *Mr. Koji's Space Dust Screen A microscope camera shows space dust on a big screen. A grain of dust that is 0.004 centimeter wide is shown 10^3 times as large. A second grain, shown 10^2 times as large, is 0.6 centimeter wide on the screen.* How wide does the first grain look on the screen? [4 centimeters / 0.4 centimeter / 40 centimeters / 0.04 centimeter]	**4 centimeters**	0.004 × 10^3 moves each digit 3 places to the left: 0.004 → 0.04 → 0.4 → 4 centimeters.
b2_s2	Set 3: Two steps: how wide is the second grain in real life? Centimeters:	**0.006** (also accepted: 0.006 centimeters, 0.006 centimeter, 0.006 cm)	The screen made it 10^2 times as large, so divide: 0.6 ÷ 10^2 moves each digit 2 places to the right: 0.06, then 0.006 centimeter.

Pre & Post sheet answer guide

Many items are open-ended. Grade for genuine understanding and complete sentences, not exact wording. Sheet: grade4-5/prepost/KCE_Math_PrePost_G4-5W27.pdf.

PRE-LESSON WARM-UP

#	Prompt	Model answer / grading note
1	Review: a) 5 + 8 × 4 b) 3 × [16 − (2 + 9)]	a) 5 + 32 = 37 b) 3 × [16 − 11] = 3 × 5 = 15
2	a) 10 × 10 b) 10 × 10 × 10 c) 10 × 10 × 10 × 10; the pattern	a) 100 b) 1,000 c) 10,000. Each extra factor of 10 adds one more zero; the number of zeros equals the number of tens multiplied.
3	26 × 10 and 26 × 100; what happens to the 2 and the 6	260 and 2,600. Each digit moves one place to the left for × 10 (2 tens become 2 hundreds, 6 ones become 6 tens) and two places for × 100. Accept "a zero is added" before the lesson.
4	Which number is 10 times as large as 4.7?	47
5	Kimani says 10 × 10 × 10 × 10 × 10 is 50	Model: I disagree. Five tens multiplied together make 100,000 (10^5). Kimani found 10 × 5, which is only 50.

POST-LESSON CHALLENGE

#	Prompt	Model answer / grading note
1	Standard form: a) 10^5 b) 7×10^3 c) $4 \times 10^4 + 6 \times 10^2$	a) 100,000 b) 7,000 c) 40,000 + 600 = 40,600
2	a) 2.36×10^2 b) 0.08×10^3 c) $5{,}400 \div 10^3$ d) $61 \div 10^2$	a) 236 b) 80 c) 5.4 d) 0.61
3	10^3 moon keychains at $1.25 and 10^2 planet magnets at $0.90	1,000 × $1.25 = $1,250 and 100 × $0.90 = $90. Total: $1,340.
4	Kirra says $3.9 \times 10^2 = 3.900$	Model: Kirra only wrote two zeros after the decimal, which does not change the value. Multiplying by 10^2 moves every digit two places to the left, so the 3 ones become 3 hundreds and the 9 tenths become 9 tens. The correct product is 390.

★ CHALLENGE

#	Prompt	Model answer / grading note
1	10^7 compared with 10^3; 1,000 × 10,000 as one power of 10	10^7 has 4 more factors of 10, so it is 10^4 = 10,000 times as large. $1{,}000 \times 10{,}000 = 10^3 \times 10^4 = 10^7$ (10,000,000), because 3 tens times 4 more tens is 7 tens multiplied, which gives 7 zeros.